Stochastic Depth Residual Network for Hyperspectral Image Classification

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***Abstract*— The convolutional neural network (CNN) is a feed-forward neural network with deep structure and convolution operation. In the hyperspectral image (HSI) classification, CNN has demonstrated excellent performance in extracting spectral and spatial information. However, the inherent complexity and high dimension of HSIs still limit the performance of most neural network models. The powerful feature extraction ability of CNN is normally achieved by dozens or more layers, which brings a series of problems such as gradient vanishing, overfitting, and slow training speed. In order to address these problems, this article presents a CNN architecture-based stochastic depth residual network (SDRN), which is specially designed for HSI data. This model takes the original 3-D cube as the input and 3-D convolution is used to extract abundant spectral and spatial features through corresponding residual blocks. In order to reduce the training time, we adopt a stochastic depth strategy. For each small batch, a sublayer is randomly discarded by an identity function. During the testing stage, the residual network with complete depth is used. Experiments on three datasets and a comparison of the state-of-art methods show that SDRN has great advantages in accuracy and training time compared with state-of-the-art HSI classification methods.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral image (HSI) classification, residual network, stochastic depth.**

I. Introduction

HYPERSPECTRAL remote sensing technology captures images in different electromagnetic wave range so as to combine both spectral and spatial information of ground objects. A hyperspectral image (HSI) normally contains tens or hundreds of bands, which can be used to accurately identify and classify surface materials for feature extraction and image classification. Based on this characteristic, hyperspectral remote sensing has been successfully applied to agriculture [1], [2], mining [3], military [4], [5], and other fields.

Many different methods have been proposed to classify HSI data, such as Bayesian classifier [6], support vector machine (SVM) [7], Gaussian maximum likelihood [8], random forest [9], sparse representation [10], [11], iterative multitask regression (IMR) [12], and logistic regression [13]. These methods require the manual design of image features in spectral, spatial, or joint spectral-spatial domains [14]–[18]. However, the intrinsic high complexity of HSIs makes the manual feature designing task a challenging problem. Moreover, in many cases, the classification method cannot be reliably generalized to new data.

On the contrary, deep-learning-based methods, such as stacked auto-encoder (SAE) [19], [20], deep belief network (DBN) [21], [22], convolutional neural network (CNN) [23]–[26], and recurrent neural networks (RNN) [27], can extract features from HSI automatically. Among these methods, both SAE and DBN are full connection neural networks. With the deepening of the network, the number of parameters of the model can be very large, which is not conducive for the training of the model and requires more computational power. In addition, although these two network models can effectively extract features of HSIs and improve the classification accuracy to some extent, they often take 1-D data as input, which makes them unable to effectively use the spatial information of HSIs. Therefore, the research community widely adopts CNN-based classification methods when both spectral and spatial information needs to be extracted.

Depending on the types of input data, convolution operations can be conducted in 1-D, 2-D, or 3-D manners. 1-D convolution is adopted when the vectorized spatial data or spectral data are used as input [28], while 2-D convolutions are normally applied to extract the band-wise or multiband spatial features [29]. In remote sensing, the same type of ground objects may show different spectral responses and shapes due to changes of illumination and weather conditions, and diversified appearance. Therefore, it is difficult to obtain highly accurate classification results using only spatial or spectral features [30]. To address this issue, joint spectral-spatial features have been widely used in recent years, which allow effective use of rich spectral information of HSIs and the spatial relationship between pixels. Therefore, the mainstream deep learning methods of HSI classification uses 3-D convolution to extract spectral-spatial features [25], [30]–[34]. However, most 3-D-CNN methods contain dozens or more layers, which

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brings a series of problems such as high computational cost, gradient vanishing, overfitting, and slow convergence speed.

To address this issue, Zhong *et al.* [35] designed an end-to-end spectral-spatial residual network (SSRN). It takes the original 3-D cube as the input and continuously extracts both spectral and spatial features through spectral and spatial residuals. Residual blocks connect other 3-D convolutional layers through identity mapping, which facilitates the back-propagation of the gradient and thus alleviates the accuracy decline of other deep learning models. However, the training of SSRN is time-consuming and a large number of training blocks may lead to overfitting. Li *et al.* [36], on the other hand, proposed a light 3-D CNN that requires much fewer parameters compared with other 3-D-CNN methods. However, the light networks may lead to a reduced accuracy compared with full networks.

Stochastic depth network [37] trains a shallow network by randomly discarding layers and restores the complete network containing all the layers in the testing process. It leads to a lightweight network but does not degrade its performance. Inspired by this model, in this article, we propose a new HSI classification model based on the stochastic depth residual network (SDRN). Different from the stochastic depth network [37], we design spectral and spatial residual blocks to extract the spectral and spatial features, which further enhance the performance. The difference with SSRN [35] is that, for the SSRN, it trains the network using the whole network. This may lead to two problems: long training and testing time and gradient dispersion. To solve these problems, inspired by stochastic depth network [37], we design a novel model, which could train the network with random depth spectral and spatial blocks. This not only reduces the training and testing time but also enhances the classification performance. The proposed network is composed of several stacked residuals. Spectral residuals and spatial residuals are used to extract spectral and spatial features of HSI data, respectively. The model convolution layer uses 3-D convolution and the residuals all contain shortcut connections. The shortcut connections perform identity mapping and their output is added to the output of the stack layer. Identifying shortcut connections adds neither additional parameters nor computational complexity. The whole network can still be trained end-to-end through root mean square prop (RMSprop) using backpropagation. Taking the HSI data cube as input, our model can learn robust spectral-spatial representation. To solve the long training time problem, the model adopts the same random depth strategy as in [37]. That is, for each small batch, a random subset of the layers is discarded and an identity function is used to bypass them. But the full network depth was restored during the test. Through this method, the training time is shortened and the classification accuracy is improved.

The rest of this article is organized as follows. Section II introduces the related work, and Section III describes the proposed method SDRN. Section IV explains the datasets and the experimental configurations. Section V gives the experiments on SDRN and some discussions. Finally, conclusions are drawn in Section VI.

II. Related Work

Deep CNNs have made significant achievements in image classification. The deep network can extract features and classify them in an end-to-end manner through convolution kernels. If the number of stacked layers is large enough, the features of lower layers can be combined to form more abstract features in higher layers, which has a great impact on the classification accuracy [38], [39]. However, with a deeper depth, in addition to gradient disappearance and gradient explosion, the network will also degenerate. Gradient disappeared or explosion problems can be solved by a deep residual learning framework (ResNet) proposed by He *et al.* [40].

A. Residual Learning

Given a shallow network, a deep network can be built by adding new layers. If these additional layers learn nothing but replicate the characteristics of the shallow layers, the operation is called identity mapping. If the design of the deep network cannot preset which layers are the redundant layers, it is desirable that these redundant layers are identity mapping layers so that the inputs and outputs of these layers are exactly the same. It is up to the network training process to determine which layers shall be the identity layers. In this case, the performance of the deep network should be at least the same as that of the shallow network, but the accuracy of the deep network on the training set may be saturated or even decreased. This cannot be interpreted as overfitting, because overfitting leads to better accuracy on the training set, rather it is due to deep network degradation.

Based on this hypothesis, residual learning is proposed to solve the degradation problem. For an accumulation layer structure (accumulated by several layers), given an input x, the characteristics learned are denoted as $H(x)$. It is expected that the network can learn residual $V(x) = H(x) - x$, so that the original learning feature is $V(x) + x$. The reason for this choice is that residual learning is easier than direct learning of original features. When the residual is 0, the heap layer only does identity mapping, so the network performance will not decrease. In fact, the residual will not be 0, which enables the heap layer to learn new features based on the input features so as to have better performance. The structure of residual learning is shown in Fig. 1.

B. Residual Units

The ResNet is based on the visual geometry group (VGG)19 [41] network with residual units. Compared with VGG, ResNet directly samples the input image with the convolution of stride of 2 and replaces the full connection layer with a global average pool layer, reducing the number of parameters. In order to maintain the complexity of the network, the size of the feature map is halved and the number of feature maps is doubled. ResNet adds shortcut connections every two layers. As shown in Fig. 1, ResNet uses two different residual units. The image on the left corresponds to a shallow network, while the image on the right corresponds to a deep network.

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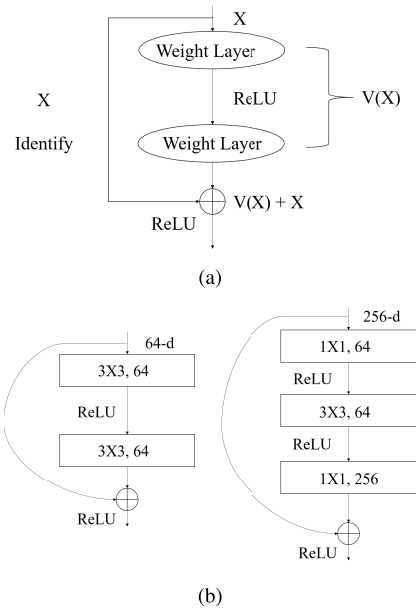


Fig. 1. (a) Residual learning: A building block. (b) Different residual units.

TABLE I
NETWORK STRUCTURE AND PARAMETERS

Data Pre-processing Module	**HSI Data (7,7,204)**	
	$CONV1$:1×1×7,24,strides (1, 1, 2)	
Spectral Feature Learning Module	k_1 **Residual Blocks**	
	F_{spe}	$CONV2_{spe}$,1×1×7,24,Input:$CONV1$
	F_{spe}	$CONV3_{spe}$,1×1×7,24,Input:$CONV2_{spe}$
	Shortcut Connection	$Add1_{spe}$,Input:$CONV1_{spe}$,$CONV3_{spe}$
	.....	
Spatial Feature Learning Module	**The First Spatial Residual Block**	
	f_{spa}	$CONV1_{spa}$,3×3×99,24,Input:$Output_{spe}$
	f_{spa}	$CONV2_{spa}$,3×3×99,24,Input:$Output_{spe}$
	F_{spa}	$CONV3_{spa}$,3×3×1,24,Input:$CONV2_{spa}$
	Shortcut Connection	$Add1_{spa}$,Input:$CONV1_{spa}$,$CONV3_{spa}$
	$k_2 - 1$ **Residual Blocks**	
	F_{spa}	$CONV4_{spa}$,3×3×1,24,Input:$Add1_{spa}$
	F_{spa}	$CONV5_{spa}$,3×3×1,24,Input:$CONV4_{spa}$
	Shortcut Connection	$Add2_{spa}$,Input:$Add1_{spa}$,$CONV5_{spa}$
	.....	
Classification Module	$Global\ Average\ Pooling$,5×5×1,Input:$Output_{spa}$	
	$Fully\ Connected$,units=16,Input:$Global\ Average\ Pooling$	

III. METHODOLOGY

In this section, we introduce the details of the proposed SDRN method, then describe the training strategy. SDRN is built based on the deep residual network. As shown in Fig. 2, the network structure consists of four modules: data pre-processing module, spectral feature learning module, spatial feature learning module, and classification module.

A. Components of 3-D CNNs

Many methods use principal component analysis (PCA) to reduce the dimension of the hyperspectral data. This produces a straightforward interface to classic CNN methods developed for color images to acquire spatial features. However, PCA distorts the spectral information of HSIs. Instead, our method uses 3-D convolution operations on images, which extracts spectral and spatial information directly without distorting the spectral information.

Given a HSI $\mathbf{X} \in \Re^{h\times w\times b}$, where h, w and b represent the height, width and number of bands of the image, respectively, our method first applies a spectral feature extraction block F_{spe}, which is composed of a 3-D convolution layer, a batch normalization (BN) layer and a rectified linear units (ReLUs) layer. As shown in Fig. 3(a), input $\mathbf{A}^c$ of 24 $r \times r \times u$ patches are convolved with q convolution kernels $\mathbf{M}^{c+1}$ of $1 \times 1 \times 7$ with zero paddings. The output $\mathbf{A}^{c+1}$ is obtained through the BN layer and the ReLU layer. The propagation rule of the function F_{spe} for the spectral feature extraction module can be expressed by

$$F_{\text{spe}}(\mathbf{A}^c) = \text{ReLU}(\text{BN}(\mathbf{A}^c * \mathbf{M}^{c+1})) \tag{1}$$

where $*$ is the convolution operation.

There are two kinds of 3-D CNN blocks for spatial information extraction as defined in functions f_{spa} and F_{spa}. f_{spa} is composed of a 3-D convolution layer, a BN layer, and a ReLU layer. As shown in Fig. 3(b), the input A^g of 24 $r \times r \times u$ patches are first convolved with $\mathbf{E}^{c+1}$ with q $3 \times 3 \times u$ filters without padding. This is followed by the BN layer and the ReLU layer to generate $\mathbf{A}^{g+1}$. The propagation rule of function f_{spa} can be expressed by

$$f_{\text{spa}}(\mathbf{A}^g) = \text{ReLU}(\text{BN}(\mathbf{A}^g * \mathbf{E}^{c+1})). \tag{2}$$

Function F_{spa} has similar structures as f_{spa} but with different filter sizes as shown in Fig. 3(c). Its propagation rule is defined as

$$F_{\text{spa}}(\mathbf{A}^j) = \text{ReLU}(\text{BN}(\mathbf{A}^j * \mathbf{R}^{c+1})). \tag{3}$$

B. Spectral-Spatial Network

As shown in Fig. 2, the SDRN is mainly composed of k_1 spectral feature learning blocks, k_2 spatial feature learning blocks, and a pooling layer and full connection layers. In our method, we first extract the spectral information. If the spatial feature extraction is carried out first, part of the spectral information will be lost. We use stochastic depth to conduct network training, whose main idea is to randomly drop some residual blocks in the training process and restore the complete network containing all residual blocks during the testing. According to [37], stochastic depth can be regarded as an implicit set of networks with different depths. That is to say, when designing the network, we should ensure that the network can carry out spectral and spatial feature extraction at least once in the case that all residual blocks are dropped. As shown in Fig. 2, a transparent residue-learning branch indicates that the residue-learning block will be dropped with a certain probability. When the residue-learning block is dropped, the network will continue to propagate from the identity mapping branch.

Table I lists the overall structure and detailed parameters of the SDRN network. The size of the input to the network is $7 \times 7 \times b$, where b is the number of bands of HSIs. Before the spectral feature learning module, a data preprocessing

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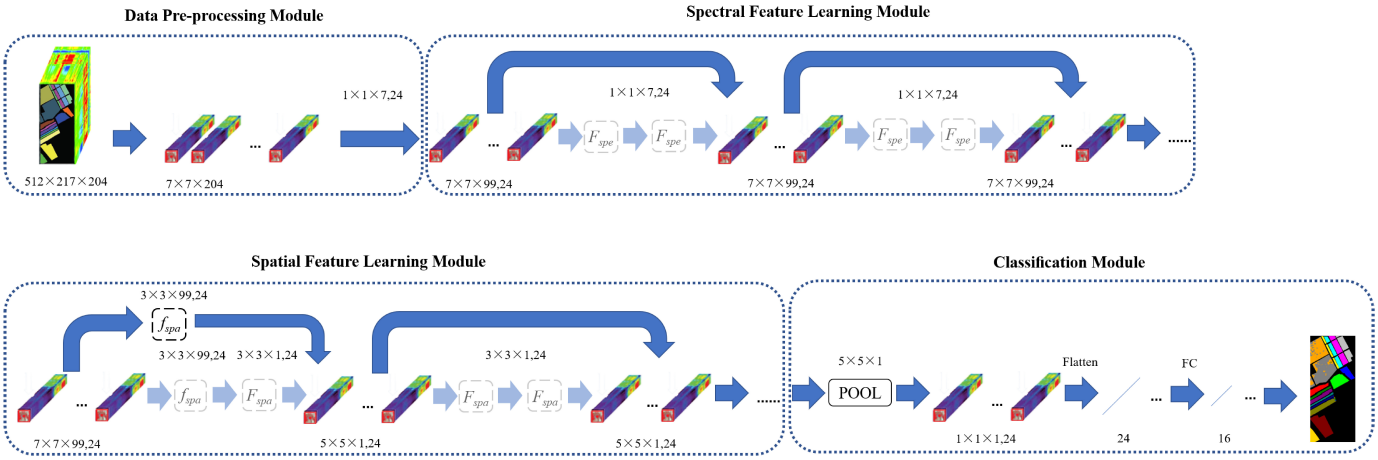


Fig. 2. SDRN.

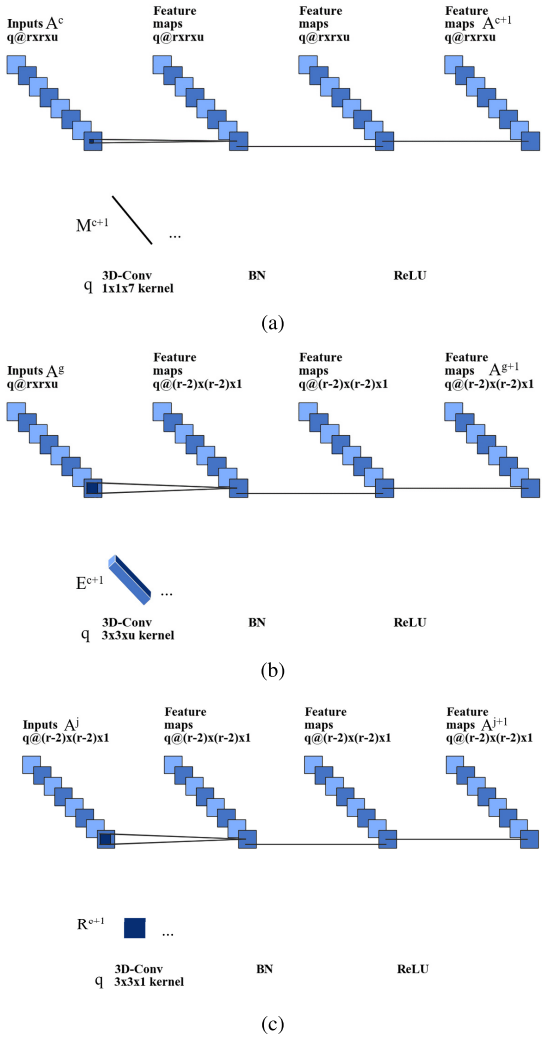


Fig. 3. (a) Spectral function F_{spe}. (b) Spatial function in the shortcut connection f_{spa}. (c) Spatial function F_{spa}.

module is used to extract features and reduce the dimension of the HSIs. Here, CONV1 is used to extract features from the spectral domain. This is also required by the stochastic depth training method whose probability of survival of the first convolution should be 1, which ensures that at least one spectral feature extraction is done when all the residual blocks are discarded.

After the data preprocessing module, there are k_1 spectral features learning blocks and k_2 spatial feature learning blocks to extract the features from spectral and spatial domains, respectively. In the spectral/spatial feature learning block, as shown in Fig. 2, there is a shortcut connection to skip the convolution process composed of spectral/spatial functions (F_{spe}, F_{spa}, or f_{spa}). The shortcut connection can be expressed by identity transformation, which keeps the origin feature. The forward propagation rule of spectral feature learning block in the training stage can be expressed as follows:

$$H_l^{\text{spe}} = \text{ReLU}\left(F_{\text{spe}}\left(F_{\text{spe}}\left(H_{l-1}^{\text{spe}}\right)\right) + id\left(H_{l-1}^{\text{spe}}\right)\right) \tag{4}$$

where the H_l^{spe} is the output of l_{th} spectral residual block and the id is an identity transformation.

To ensure the spatial feature extraction is processed at least once even all the residual blocks are skipped, the identity mapping branch of the first spatial residual block is changed to a function f_{spa}. Therefore, the forward propagation formula of the first residual block in the spatial feature learning module is calculated as follows:

$$H_1^{\text{spa}} = \text{ReLU}\left(F_{\text{spa}}\left(f_{\text{spa}}\left(H_0^{\text{spa}}\right)\right) + f_{\text{spa}}\left(H_0^{\text{spa}}\right)\right) \tag{5}$$

where the H_1^{spa} is used to represent the output of the first spatial residual block and H_0^{spa} is the output of spectral feature extraction.

In the next $k_2 - 1$ spatial feature extraction residual blocks, the residual branch is composed of two F_{spa}, and the identity mapping branch without functions f_{spa}. The formula is as follows:

$$H_l^{\text{spa}} = \text{ReLU}\left(F_{\text{spa}}\left(F_{\text{spa}}\left(H_{l-1}^{\text{spa}}\right)\right) + id\left(H_{l-1}^{\text{spa}}\right)\right) \tag{6}$$

where the H_l^{spa} is used to represent the output of the l_{th} spatial residual block and H_{l-1}^{spa} is the output of $(l-1)_{\text{th}}$ output of spatial features.

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Given the spectral and spatial feature learning blocks, the forward propagation formula of the whole feature learning module is summarized as follows:

$$
\begin{aligned}
H_l^e &= \mathrm{ReLU}\big(F_1\big(F_2\big(H_{l-1}^e\big)\big) + id\big(H_{l-1}^e\big)\big) \\
F_1 &= \begin{cases} F_{\mathrm{spe}}, & \text{if } e \text{ is spe} \\ F_{\mathrm{spa}}, & \text{if } e \text{ is spa} \end{cases} \\
F_2 &= \begin{cases} F_{\mathrm{spe}}, & \text{if } e \text{ is spe} \\ f_{\mathrm{spa}}, & \text{if } e \text{ is spa and } l \text{ is 1} \\ F_{\mathrm{spe}}, & \text{if } e \text{ is spa and } l \text{ is not 1} \end{cases} \\
id &= \begin{cases} id, & \text{if } e \text{ is spe} \\ f_{\mathrm{spa}}, & \text{if } e \text{ is spa and } l \text{ is 1} \\ id, & \text{if } e \text{ is spa and } l \text{ is not 1.} \end{cases}
\end{aligned} \tag{7}
$$

After the spectral and spatial feature learning blocks, 24 feature cubes with the size of $5 \times 5 \times 1$ are extracted and fed into the pooling layer to obtain 24 feature cubes with the size of $1 \times 1 \times 1$. Then, the feature cube is flattened and transformed into feature vectors. At last the fully connected layer, according to the number of categories, generate an output vector $\hat{y} = [\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_{16}]$.

C. Training Strategy

As the network goes deeper, it takes a longer time to train the network. The efficiency of forward propagation and backpropagation is linearly related to the depth of the network. A shallow network allows information to flow quickly forward and backward, so it can be effectively trained in a short period of time. However, with the redundant spectral and spatial information of HSIs, it is insufficient for a shallow network to extract and express the information. Deep networks have sufficient model complexity, yet they are difficult to train due to gradient vanish and network degenerates and take a lot of time.

To solve this problem, inspired by the stochastic depth model [37], we could define a network structure that obtains a shallow network during the training process and a deep network with strong expression ability during the testing process. This can be achieved by building a deep network composed of spectral and spatial feature learning blocks, randomly discarding some blocks during the training process, and restoring the complete network containing all blocks during the testing process. In this way, the training time can be significantly shortened because no forward and backward propagations are needed for the discarded blocks. Meanwhile, it is found that the ResNet network usually contains redundant layers [37]. So randomly discarding blocks is similar to the classic dropout neurons during the training process, which may enhance the performance of the network. The purpose of the dropout neurons is to avoid overfitting. In our proposed method, the process of randomly discarding blocks is similar to the dropout process. We conducted experiments to evaluate whether dropout neurons could enhance the performance. Moreover, when the network is deep, the strength of signals may be too weak to effectively the deep layers. By dropping some of the feature learning blocks, high-level blocks can receive more information from the shallow blocks for sufficient training, so the model can have better expressive ability.

In implementing this algorithm, we set a Bernoulli random variable $b_l \in \{0, 1\}$, which means that the l_{th} feature learning block works normally when $b_l = 1$ or is discarded when $b_l = 0$. Therefore, the forward propagation rule (7) becomes

$$
H_l^e = \mathrm{ReLU}\big(b_l F_1\big(F_2\big(H_{l-1}^e\big)\big) + id\big(H_{l-1}^e\big)\big) \tag{8}
$$

where b_l is multiplied with the function $F_1(F_2(H_{l-1}^e))$ of the l_{th} residual. When $b_l = 0$, the residual becomes the identity function, such that

$$
H_l = id(H_{l-1}). \tag{9}
$$

We adopt the same linear attenuation survival probability setting method as [37]. Let P_l represent the survival probability of the l_{th} residual block, and P_L represent the survival probability of the last residual block, constant set to 0.5. The survival probability of the l_{th} residual block is a function of l

$$
P_l = 1 - \frac{l(1 - P_L)}{L}. \tag{10}
$$

Due to the stochastic strategies, inactivation of a part of the block actually realizes an implicit model ensemble. With the random depth of the model during training and the determination of the depth of the model during prediction, the models of different depths are actually integrated during testing. In this way, the training of random depth performs the implicit model fusion of different numbers of remaining spectral and spatial features. This can solve the problem of the redundancy layer in the model. Moreover, the spatial feature residuals in the deeper layers can extract spatial information directly from spectral feature residuals in random shallower layers, which effectively overcomes overfitting and increases the model generalization ability.

The training time of applying stochastic depth no longer increases with the depth increase of network but depends on the expected depth of the network. During training, the probability that the l_{th} residual block being skipped is $(1 - P_l)$. Then the number of effective residuals $\tilde{L}$ during training becomes a random variable, and its expectation is given by

$$
E(\tilde{L}) = \sum_{l=1}^{L} P_l. \tag{11}
$$

Under the rules of linear attenuation and P_L equals 0.5, the number of effective residual block during training is $E(\tilde{L}) = (3L - 1)/4$. If L is large, it is approximately $E(\tilde{L}) = 3\ L/4$. This decrease in depth further alleviates the gradient vanishing problem. Under the linear attenuation rule and $P_L = 0.5$, approximately 25% training time can be saved.

IV. Experiments

Three hyperspectral datasets are used to illustrate the effectiveness of the proposed method: Indian Pines, Pavia University, and Salinas. In the experiments, the overall accuracy (OA), the average accuracy (AA), and the kappa coefficient (kappa) are used to evaluate the performance. The experiments are run ten times with randomly selected training

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TABLE II
TRAINING, VALIDATION, AND TESTING DATA IN INDIAN PINES DATASET

#	Class	Training	Val	Test
1	Alfalfa	10	1	35
2	Corn-notill	286	131	1011
3	Corn-mintill	166	83	581
4	Corn	48	22	167
5	Grass-pasture	97	42	344
6	Grass-trees	146	69	515
7	Grass-pasture-mowed	6	3	19
8	Hay-windrowed	96	55	327
9	Oats	4	4	12
10	Soybean-notill	195	94	683
11	Soybean-mintill	491	264	1700
12	Soybean-clean	119	56	418
13	Wheat	41	26	138
14	Woods	253	136	876
15	Buildings-Grass-Trees-Drives	78	34	274
16	Stone-Steel-Towers	19	5	69
-	Total	2055	1025	7169

TABLE III
TRAINING, VALIDATION, AND TESTING DATA IN PAVIA UNIVERSITY DATASET

#	Class	Training	Val	Test
1	Asphalt	664	670	5297
2	Meadows	1865	1810	14974
3	Gravel	210	241	1648
4	Trees	307	333	2424
5	Painted metal sheets	135	134	1076
6	Bare Soil	503	500	4026
7	Bitumen	133	133	1064
8	Self-Blocking Bricks	369	363	2950
9	Shadows	95	97	755
-	Total	4281	4281	34214

and testing data using the region sampling strategy [42] and the random sampling strategy. The mean results and standard deviation of the metrics are reported.

A. Hyperspectral Datasets

Indian Pines dataset was acquired in 1992 using the airborne visible infrared imaging spectrometer (AVIRIS) in Indiana, the United States. The image was cut to the size of 145×145 pixels. The AVIRIS sensor captures images in 0.4–2.5 μm with 220 consecutive bands. Since 104–108, 150–163, and 220 bands are normally removed due to water absorption, only the remaining 200 bands were normally used for experiments. There are 16 categories in total, and the distribution of samples in each category is extremely unbalanced.

Pavia University dataset was captured over the Italian city of Pavia in 2003. The imaging sensor continuously captured 115 bands from 0.43 to 0.86 μm, resulting in images with a spatial resolution of 1.3 m. Among them, 12 bands are removed due to the influence of noise, with the remaining 103 spectral bands used for experiments. The size of the data is 610×340, so it contains a total of 207 400 pixels. However, most of them are background pixels, with only 42 776 pixels being foreground in nine categories, including trees, Asphalt, Bricks, and Meadows.

Salinas dataset was also taken by the AVIRIS imaging spectrometer. It was captured over the Salinas valley in CA, USA, with a spatial resolution of 3.7 m. Similar to Indian Pines, water absorption bands are removed, leaving 204 bands for experiments. The image size is 512×217, which 54 129 foreground pixels in 16 classes and 56 975 background pixels.

TABLE IV
TRAINING, VALIDATION, AND TESTING DATA IN SALINAS DATASET

#	Class	Training	Val	Test
1	Brocoli_green_weeds_1	201	181	1627
2	Brocoli_green_weeds_2	373	356	2997
3	Fallow	198	183	1595
4	Fallow_rough_plow	140	146	1108
5	Fallow_smooth	268	302	2108
6	Stubble	396	372	3191
7	Celery	358	358	2863
8	Grapes_untrained	1128	1129	9014
9	Soil_vinyard_develop	621	610	4972
10	Corn_senesced_green_weeds	328	314	2636
11	Lettuce_romaine_4wk	107	124	837
12	Lettuce_romaine_5wk	193	199	1535
13	Lettuce_romaine_6wk	92	95	729
14	Lettuce_romaine_7wk	107	105	858
15	Vinyard_untrained	727	759	5782
16	Vinyard_vertical_trellis	181	185	1441
-	Total	5418	5418	43293

TABLE V
OA (%) OF THE PROPOSED METHOD WITH DIFFERENT INPUT SIZES

Spatial Size	INDIAN PINES	PAVIA UNIVERSITY	SALINAS
3×3	72.51±4.69	89.34±2.51	91.06±1.71
5×5	74.44±4.56	89.59±3.70	91.31±1.89
7×7	75.11±3.48	91.56±2.07	92.53±1.13
9×9	75.62±5.95	91.08±2.19	93.02±1.21

TABLE VI
OA (%) OF THE PROPOSED METHOD WITH DIFFERENT REGULARIZERS

Proposed	INDIAN PINES	PAVIA UNIVERSITY	SALINAS
None	71.52±3.28	87.94±1.83	91.46±1.76
Dropout	72.50±4.66	88.02±2.50	91.75±1.32
BN	72.48±4.96	90.79±2.95	92.24±1.50
Both	75.11±3.48	91.56±2.07	92.53±1.13

Each dataset is divided into three subsets for training, validation, and testing. The training set is used to train models under comparison. The validation set is used to evaluate the performance of the trained model in each iteration. The final optimal trained models are evaluated on the testing set and compared with each other. Tables II–IV list the specific number of samples selected for training, validation, and testing in each category, respectively.

B. Experimental Configuration

For data preprocessing, we divided the original HSI into small 3-D blocks of images and built the sample set. Given a HSI block of d channels, a window of $h \times h$ was used to traverse the block sample with each pixel as the center, with cubes being extracted with zero-padding at the boundary. Suppose an HSI X contains N classifiable pixels $\mathbf{X} = \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N \in R^{1\times1\times d}$, $\mathbf{Y} = \mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N \in R^{1\times1\times1}$ are the corresponding label, where d and l are the number of

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TABLE VII

CLASSIFICATION RESULTS ON INDIAN PINES DATASET USING REGION BASED SAMPLING METHOD

Class	MLP	SVM-RBF	SAE	3D-CNN	3D-DCNN	M3D-DCNN	DR-CNN	SSRN	Proposed
1	80.55±9.88	77.77±11.78	47.17±12.28	26.34±20.50	75.43±21.37	31.20±4.81	82.40±22.94	0.00±0.00	**98.95±1.47**
2	63.98±5.46	56.12±5.01	37.30±4.33	61.59±4.30	56.24±6.74	31.27±1.10	44.10±8.56	**66.17±11.86**	64.63±7.25
3	50.10±8.79	53.36±11.37	31.89±2.38	55.94±8.68	56.91±10.19	29.04±8.28	59.78±17.46	66.86±9.14	**69.03±14.83**
4	62.65±8.48	65.84±5.17	26.06±4.99	65.44±4.73	41.50±14.36	18.19±0.82	51.59±3.73	57.55±35.88	**90.35±8.23**
5	60.22±3.26	64.80±9.57	45.08±4.68	73.13±13.74	69.44±18.86	48.80±8.29	67.23±21.64	73.00±20.59	**74.76±14.75**
6	87.50±3.39	86.58±2.37	78.95±4.34	**94.07±4.39**	93.46±4.43	84.95±8.25	92.63±6.79	89.48±6.77	91.55±6.45
7	54.54±18.55	81.81±3.71	17.26±15.45	0.00±0.00	95.65±6.14	48.28±10.59	**100.00±0.00**	0.00±0.00	95.65±6.14
8	82.63±5.34	84.11±6.46	86.49±5.01	**99.29±0.99**	82.18±13.07	94.29±3.70	93.71±6.94	98.49±1.71	94.49±7.78
9	22.91±2.94	64.58±21.24	34.02±28.07	0.00±0.00	92.85±10.10	78.08±19.88	**95.83±2.94**	0.00±0.00	71.90±26.86
10	64.09±9.04	**67.05±12.87**	46.84±19.65	45.56±7.10	65.97±13.01	29.95±15.75	59.71±5.82	59.40±11.69	49.04±7.81
11	56.90±7.48	65.17±4.37	62.41±1.82	74.53±6.17	59.15±2.37	66.28±6.86	58.04±8.67	62.48±13.59	**74.73±3.85**
12	39.02±10.67	43.38±14.42	15.88±2.09	43.30±2.59	34.54±5.95	18.53±1.64	29.11±5.13	47.65±8.89	**52.12±22.54**
13	97.35±1.25	**97.35±1.25**	88.49±9.23	94.91±2.75	91.96±7.42	91.59±10.41	93.90±8.19	93.58±7.02	96.33±3.26
14	90.57±6.96	91.33±7.93	79.42±2.06	95.94±5.18	87.40±4.20	75.24±6.92	88.01±5.79	**97.81±0.55**	94.88±6.13
15	51.02±2.85	47.35±2.01	58.02±7.21	71.32±7.21	46.09±11.40	34.19±4.69	63.28±7.00	49.73±12.83	**80.23±12.43**
16	79.84±3.82	84.31±4.40	93.26±6.65	96.41±2.90	97.07±3.11	83.09±6.87	**97.74±3.18**	63.70±45.06	96.41±5.07
OA(%)	65.64±2.93	67.75±2.40	55.11±1.24	71.48±2.16	65.58±4.13	53.21±2.43	64.41±5.86	70.09±4.21	**75.11±3.48**
AA(%)	65.24±0.36	70.68±1.82	53.03±3.77	62.36±1.89	71.62±3.18	53.93±0.66	73.57±4.49	57.87±6.41	**80.94±1.78**
Kappa(%)	60.94±3.24	63.31±2.76	48.75±1.19	67.14±2.35	61.00±4.72	46.12±2.50	59.74±6.50	66.01±4.67	**71.56±3.89**

TABLE VIII

CLASSIFICATION RESULTS ON PAVIA UNIVERSITY DATASET USING REGION BASED SAMPLING METHOD

Class	MLP	SVM-RBF	SAE	3D-CNN	3D-DCNN	M3D-DCNN	DR-CNN	SSRN	Proposed
1	85.64±5.85	82.79±8.05	65.63±2.36	**92.73±4.40**	85.57±6.74	87.97±8.15	82.05±10.83	83.47±5.60	88.59±5.64
2	73.46±3.44	71.00±7.19	87.89±2.92	**95.00±2.34**	75.18±8.07	88.95±6.73	82.34±5.86	94.50±2.73	91.42±3.75
3	51.56±9.82	59.06±12.21	38.60±3.01	11.73±12.29	73.38±9.52	17.44±14.60	69.61±7.44	75.26±15.66	**82.40±4.65**
4	76.84±15.03	93.82±3.25	70.04±3.72	90.93±2.24	93.30±4.23	64.02±29.26	96.36±2.18	96.58±1.61	**98.47±0.43**
5	99.36±0.33	99.77±0.07	98.43±1.21	93.98±4.60	99.78±0.11	98.45±1.80	98.53±1.09	100.00±0.00	**100.00±0.00**
6	67.64±12.73	67.60±12.52	51.73±5.06	35.78±13.95	67.86±13.78	33.92±22.56	91.76±1.44	91.92±4.25	**96.32±3.00**
7	54.13±20.12	75.52±15.72	41.82±9.45	17.23±24.37	88.90±4.83	24.19±34.21	90.72±1.53	98.29±1.08	**98.42±2.03**
8	86.45±3.87	85.17±6.36	48.59±11.75	83.12±10.28	90.30±3.05	88.29±5.46	**94.89±2.81**	80.18±16.84	82.92±2.48
9	99.68±0.19	99.13±0.45	64.84±3.01	96.01±3.44	**99.91±0.06**	97.11±2.23	99.33±0.45	99.64±0.40	99.69±0.43
OA(%)	75.74±3.01	76.36±3.93	67.13±2.04	79.87±2.91	80.20±2.96	75.42±3.38	86.01±4.27	90.87±2.61	**91.56±2.07**
AA(%)	77.20±1.25	81.54±2.97	63.06±1.84	68.50±3.67	86.02±1.43	66.71±8.77	89.51±2.43	91.09±1.09	**93.13±1.21**
Kappa(%)	68.96±3.68	70.13±4.73	58.11±1.66	72.63±3.92	74.65±3.39	66.49±5.38	82.07±5.25	88.03±3.38	**89.02±2.64**

TABLE IX

CLASSIFICATION RESULTS ON SALINAS DATASET USING REGION BASED SAMPLING METHOD

Class	MLP	SVM-RBF	SAE	3D-CNN	3D-DCNN	M3D-DCNN	DR-CNN	SSRN	Proposed
1	98.06±0.57	98.96±0.29	84.95±4.38	99.29±0.99	98.45±2.18	92.86±1.63	**100.00±0.00**	99.89±0.10	99.75±0.18
2	99.51±0.33	98.45±0.41	88.01±5.75	100.00±0.00	99.69±0.36	99.92±0.11	99.73±0.21	100.00±0.00	**100.00±0.00**
3	75.45±11.00	78.12±19.63	83.90±4.89	**99.37±0.62**	95.35±6.08	84.81±7.05	88.95±5.15	98.84±1.62	95.34±3.98
4	98.85±0.19	97.68±1.55	59.05±10.84	98.05±1.61	99.21±0.05	99.49±0.29	99.33±0.22	**99.84±0.15**	99.70±0.30
5	**98.47±0.46**	91.92±3.94	91.54±6.27	97.96±0.08	93.77±2.28	93.66±3.98	96.47±3.00	92.02±4.91	97.47±1.11
6	99.53±0.03	98.19±1.34	94.19±1.94	98.91±0.84	98.26±1.77	99.13±1.14	99.78±0.28	**99.84±0.11**	99.79±0.25
7	99.73±0.08	99.49±0.19	90.29±4.02	99.82±0.22	99.66±0.27	99.23±0.20	99.58±0.14	99.98±0.01	**99.98±0.01**
8	84.59±9.99	78.97±8.90	70.06±7.13	84.70±4.36	80.51±0.18	83.69±3.88	**88.61±3.23**	85.14±1.60	87.49±3.02
9	99.65±0.16	97.38±1.77	95.87±2.40	**100.00±0.00**	98.86±0.71	96.80±2.54	98.19±1.57	99.84±0.10	99.80±0.27
10	85.55±4.57	87.08±1.72	90.10±8.07	92.24±0.51	91.11±2.85	87.13±4.12	91.11±4.88	95.06±0.67	**95.56±1.87**
11	87.37±0.88	89.14±0.86	98.10±1.33	93.77±3.77	94.62±1.21	77.44±8.31	96.70±2.89	98.91±0.69	**99.18±0.28**
12	99.71±0.18	97.25±2.31	87.26±7.03	97.18±2.82	99.87±0.18	93.25±7.09	99.69±0.43	99.65±0.43	**99.95±0.06**
13	97.45±3.09	98.90±1.03	82.56±9.10	95.64±2.71	99.90±0.12	99.86±0.19	99.59±0.57	99.95±0.06	**100.00±0.00**
14	85.87±1.40	86.63±2.84	83.09±11.78	88.59±7.75	92.22±1.24	91.88±0.63	**98.51±0.55**	94.39±2.36	93.90±1.21
15	41.36±12.98	56.56±3.53	58.74±5.09	61.42±7.08	**74.55±4.29**	63.09±5.35	62.19±11.60	72.17±4.73	70.20±7.21
16	82.86±12.05	85.09±7.90	94.46±3.85	96.36±2.72	96.43±0.51	91.72±1.67	96.84±2.20	96.67±2.72	**97.27±1.75**
OA(%)	85.69±1.13	86.04±1.14	81.11±2.06	90.20±1.99	90.73±0.28	88.08±0.46	90.93±1.06	92.12±1.27	**92.53±1.13**
AA(%)	89.63±0.80	89.99±1.01	84.51±2.50	93.95±1.40	94.53±0.35	90.87±0.42	94.70±0.42	95.76±0.92	**95.96±0.38**
Kappa(%)	84.00±1.22	84.41±1.25	79.02±2.32	89.07±2.23	89.69±0.32	86.72±0.53	89.88±1.19	91.22±1.41	**91.67±1.27**

available spectral bands and the number of land categories of the dataset, respectively. The extracted cubes forms a new dataset $\mathbf{Z} = \mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{Z}_N \in R^{h \times h \times d}$.

Before training the model, some factors should be considered. We mainly discuss three factors: sampling strategy, the size of the input data cube, and the network regularization methods. For the sampling strategy, according to Liang *et al.* [42], the traditional random sampling method can lead to overlapped training and testing samples, which means the testing samples may be explicitly used during the training stage. Fig. 4 gives such an example when a 5 × 5 sliding window is used for feature extraction. The gray color region

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TABLE X
CLASSIFICATION RESULTS ON INDIAN PINES DATASET WITH RANDOM SAMPLING METHOD

Class	MLP	SVM-RBF	SAE	3D-CNN	3D-DCNN	M3D-DCNN	DR-CNN	SSRN	Proposed
1	11.11±2.26	54.62±6.92	95.83±2.94	66.02±15.23	78.76±5.90	80.98±10.12	97.22±2.26	93.59±5.66	**97.84±1.52**
2	77.75±2.51	81.11±0.93	76.65±0.87	98.36±0.83	96.61±0.86	96.28±0.38	98.62±0.75	98.79±0.70	**99.09±0.78**
3	55.17±1.06	69.62±1.78	73.74±0.81	98.20±0.18	93.20±1.25	95.77±1.42	99.44±0.18	98.05±1.33	**99.70±0.29**
4	52.55±4.32	70.19±1.08	83.21±5.57	96.99±1.81	88.98±2.93	91.51±1.53	99.64±0.24	98.46±1.73	**100.00±0.00**
5	86.26±1.17	91.96±1.28	86.15±1.23	97.46±1.73	96.10±2.61	97.85±1.34	98.87±0.85	98.42±1.33	**99.90±0.13**
6	94.34±1.81	95.94±1.62	89.73±1.12	99.67±0.09	99.34±0.45	99.60±0.27	99.77±0.21	**99.94±0.08**	99.67±0.18
7	62.12±2.14	80.30±5.66	**97.22±3.92**	41.84±21.80	71.05±21.16	76.14±14.28	96.96±4.28	64.54±16.17	94.91±4.08
8	99.38±0.44	98.42±0.56	93.72±1.07	100.00±0.00	98.66±0.94	98.66±0.95	100.00±0.00	100.0±0.00	**100.0±0.00**
9	0.00±0.00	58.33±15.5	77.97±8.57	0.00±0.00	89.33±5.31	95.39±3.26	95.83±2.94	4.71±9.92	**100.00±0.00**
10	67.13±2.31	75.97±3.25	87.60±2.94	96.84±1.34	94.54±0.33	96.55±0.84	98.28±0.36	98.23±0.88	**98.55±0.88**
11	80.88±0.66	87.03±0.86	86.03±0.62	98.24±0.59	97.68±0.40	98.24±0.66	**99.69±0.14**	99.29±0.40	99.38±0.59
12	63.92±3.06	80.94±1.10	74.36±0.32	97.38±1.34	94.29±0.16	95.31±0.99	98.94±0.68	98.38±1.05	**99.44±0.22**
13	98.37±0.28	98.17±1.31	97.22±1.49	100.00±0.00	99.06±0.34	100.00±0.00	**100.00±0.00**	99.92±0.21	99.30±0.58
14	96.21±1.11	96.93±0.85	95.26±0.63	99.62±0.14	97.70±1.39	99.24±0.66	99.96±0.04	99.56±0.45	**100.00±0.00**
15	57.90±0.93	61.25±1.19	84.39±2.78	98.38±1.55	94.43±2.93	95.05±1.43	99.02±0.26	99.70±0.39	**100.00±0.00**
16	91.44±3.87	94.59±2.20	65.16±46.08	99.49±0.71	100.00±0.00	**100.00±0.00**	99.54±0.63	99.84±0.45	99.49±0.71
OA(%)	78.46±0.16	84.73±0.29	85.13±0.06	97.93±0.12	96.30±0.14	97.28±0.52	99.31±0.13	98.75±0.19	**99.46±0.15**
AA(%)	68.41±0.20	80.96±1.77	85.27±3.02	86.78±1.59	93.11±1.27	94.79±1.24	98.86±0.44	90.71±1.05	**99.20±0.39**
Kappa(%)	75.27±0.20	82.52±0.33	83.01±0.09	97.36±0.14	95.78±0.16	96.90±0.59	99.22±0.15	98.57±0.22	**99.38±0.17**

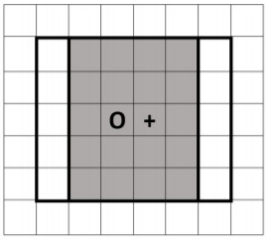


Fig. 4. 5 × 5 sliding window.

shows the overlap between the neighborhood from a training sample (O) and a testing sample (+), which means 4/5 of the 5 × 5 sliding window from the test sample has been used during the training stage. To address this problem, we adopted a region-based sampling strategy [42] for a comparison with other methods. Tables VII–IX show the results of region-based sampling strategy. To compare with other literature woks, we also conducted experiments on Indian Pine dataset using the traditional random sampling strategy, whose results are shown in Table X. From these results, we can see that the performance of the random sampling strategy is much saturated compared with the region-based sampling strategy. This is because the region-based sampling strategy overcomes the overlapping problem. For both sampling strategies, our proposed method produces the best performance, which proves the effectiveness and advantages of our method.

To analyze the influence of the size of the input data cube, we calculated the OA on three datasets using different sizes of the input data cube, i.e., 3 × 3, 5 × 5, 7 × 7, and 9 × 9. It can be seen from Table V that the size of 7 × 7 produces the better OA. Therefore, we set the size of the input data cube as 7 × 7 in the following experiments.

For the network regularization methods, we used different methods to regularize our network including BN and dropout. From Table VI, we can see that BN and dropout strategy produces the best performance. Therefore, in the experiments, we adopted the BN and dropout strategy.

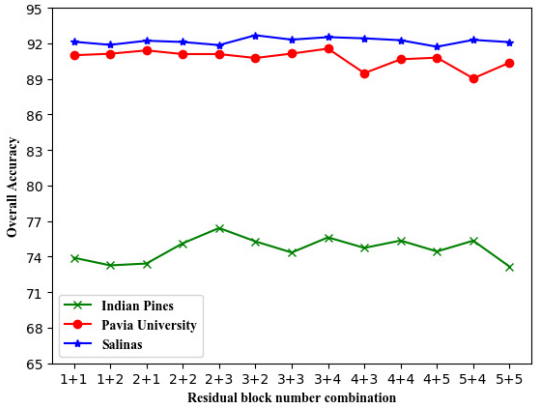


Fig. 5. OA with different numbers of spectral and spatial residual blocks. $k_1 + k_2$ means the proposed network consists of k_1 spectral residual blocks and k_2 spatial residual blocks.

In our proposed method, the amount of residual blocks determines the depth of the network. There are two kinds of residual blocks in our method, spectral and spatial residual blocks. Therefore, the number of spectral and spatial feature learning blocks (k_1 and k_2) also affects the experimental results. In order to verify the influence k_1 and k_2, we tuned the number of two residual blocks in SDRN. As shown in Fig. 5, the combinations of residual blocks are from (1, 1) to (5, 5), respectively. From Fig. 5, we can see that the performance drops when the number of blocks is larger than (3, 4). This is because a deeper network may lead to overfitting in training. A moderate number of spectral-spatial learning blocks not only can avoid overfitting but also discards useless features. From Fig. 5, we can also see that compared with other datasets, the performance on the India Pines dataset is poor. This may due to two reasons. First, compared with the other two datasets, the India Pines dataset has very few samples in several categories, so the samples are extremely unbalanced. This could severely affect the performance. Second, we adopted a different sampling strategy, i.e., the region-based sampling

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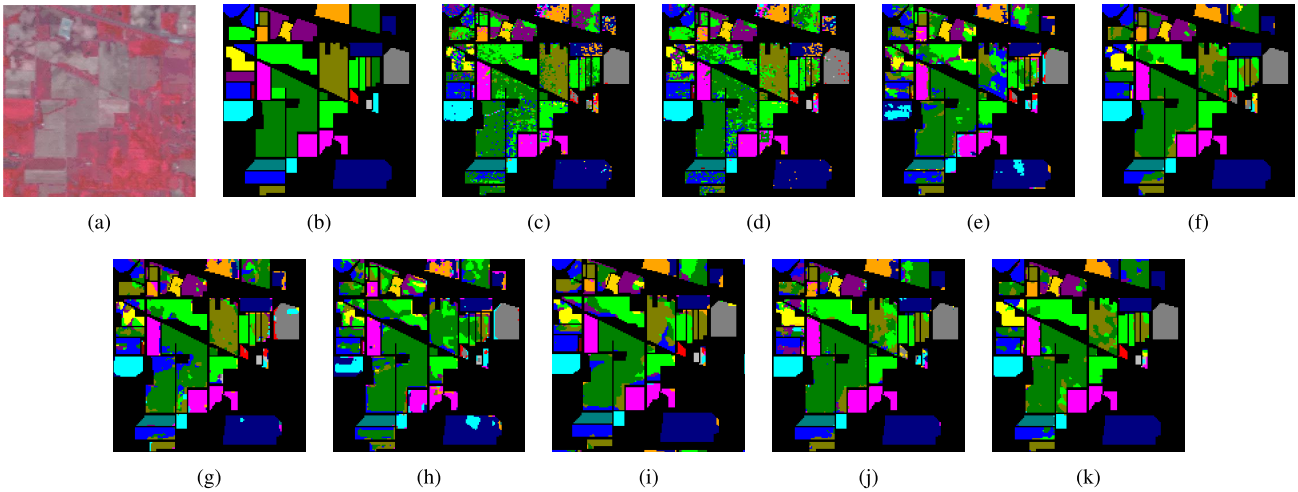


Fig. 6. Illustration of classification results on Indian Pines dataset. (a) False color image. (b) Ground truth. (c) MLP (67.78%). (d) SVM-RBF (69.98%). (e) SAE (56.70%). (f) 3-D-CNN (74.54%). (g) 3-D-DCNN (71.38%). (h) M3D-DCNN (55.63%). (i) DR-CNN (64.01%). (j) SSRN (72.63%). (k) SDRN (79.31%).

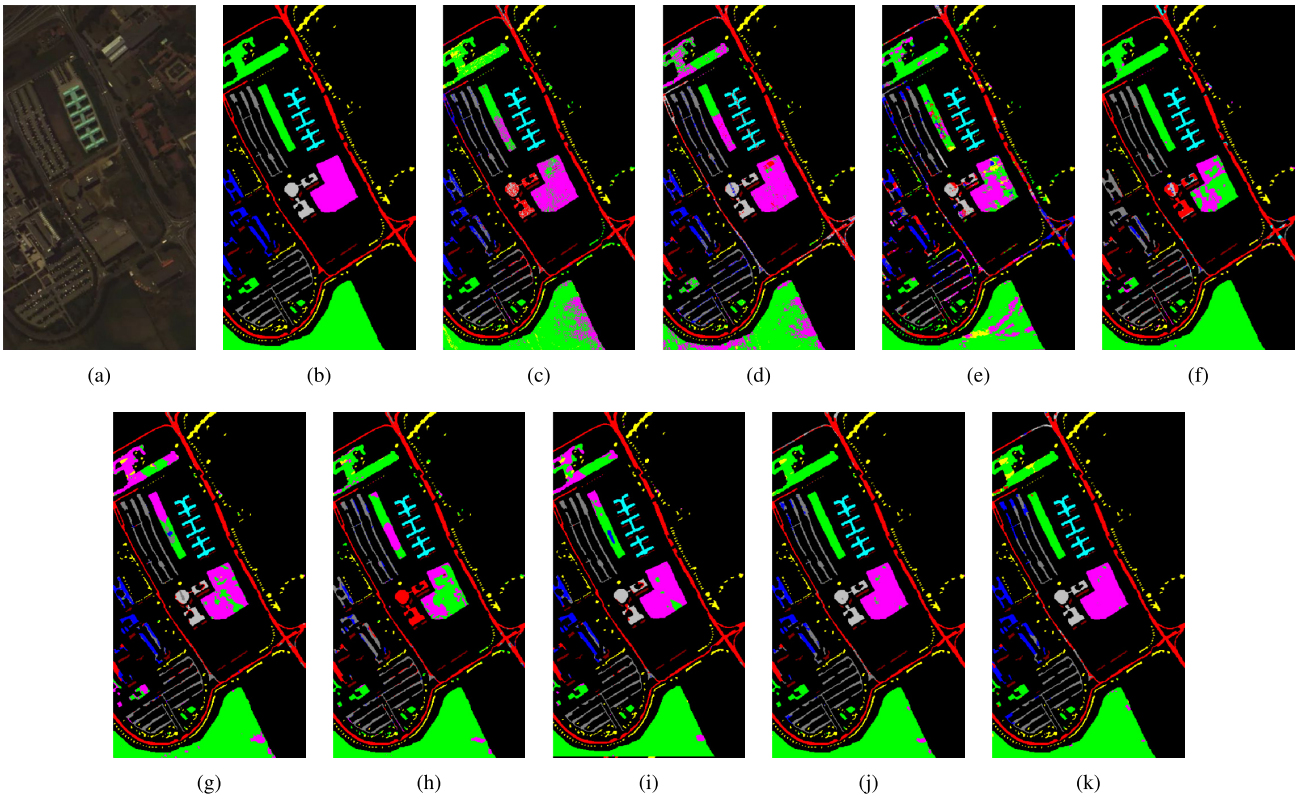


Fig. 7. Illustration of classification results on Pavia University dataset. (a) False color image. (b) Ground truth. (c) MLP (81.22%). (d) SVM-RBF (75.63%). (e) SAE (69.87%). (f) 3-D-CNN (82.44%). (g) 3-D-DCNN (82.47%). (h) M3D-DCNN (79.30%). (i) DR-CNN (89.84%). (j) SSRN (93.51%). (k) SDRN (94.50%).

strategy, which also affected the performance, especially for the categories with limited samples. It is very difficult to obtain the optimal parameters to ensure that the model achieves the best results on all datasets. It can be seen from Fig. 5 when $k_1 = 3$ and $k_2 = 4$, we got the best performance for the Pavia University dataset, and the second-best performance for the Salinas and Indian Pines datasets, with the best average performance for three datasets. Therefore, we set $k_1 = 3$ and $k_2 = 4$ in our experiments.

V. Discussion

To illustrate the effectiveness of our proposed method, SDRN is compared with eight different HSI classification methods. They are multilayer perceptron (MLP), SVM based on radial basis function (RBF) kernel [43], SAE [19], 3-D-CNN [44], 3-D-deep convolutional neural network (DCNN) (M3D-DCNN without multiscale) [34], diverse region-based (DR)-CNN [25], M3D-DCNN [34], and SSRN [35]. The size of the input cube for all classification methods is

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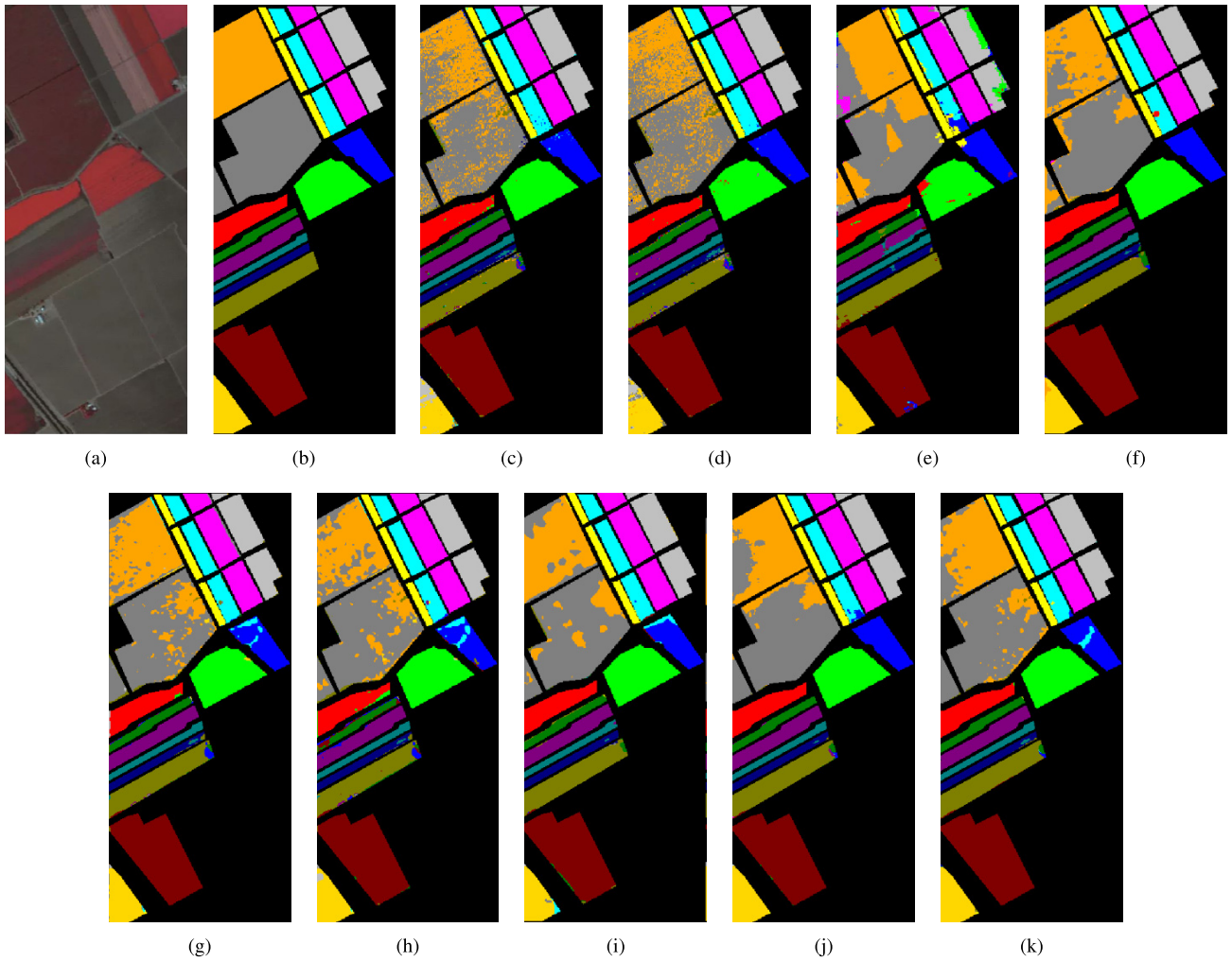


Fig. 8. Illustration of classification results on Salinas dataset. (a) False color image. (b) Ground truth. (c) MLP (87.07%). (d) SVM-RBF (87.44%). (e) SAE (83.85%). (f) 3-D-CNN (91.64%). (g) 3-D-DCNN (91.12%). (h) M3D-DCNN (88.59%). (i) DR-CNN (92.38%). (j) SSRN (93.30%). (k) SDRN (93.52%).

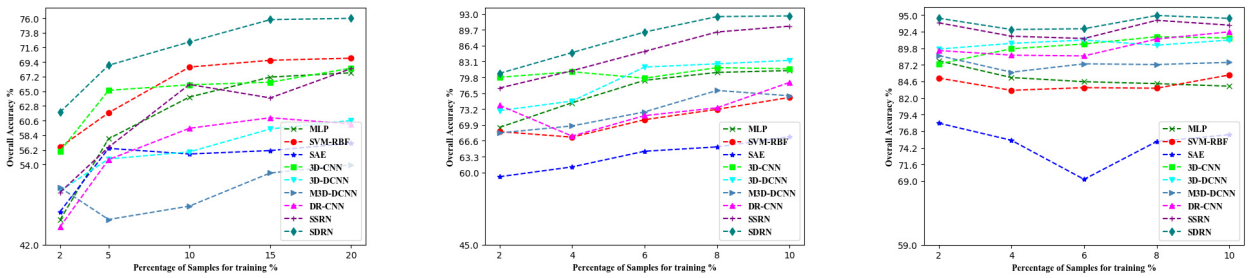


Fig. 9. OA(%) of different methods with different training percentages of samples. (a) Indian Pines. (b) Pavia university. (c) Salinas.

$7 \times 7 \times d$, where d represents the number of available bands for HSIs. All the classification methods were tuned to the best settings. The training, validation, and testing data are divided the same as Tables II–IV.

Tables VII–IX show the OA, AA, and Kappa coefficients of all classification methods on the three datasets, including the classification accuracy of each category. From the tables, we can see that the proposed SDRN has obtained the best performance among all methods on all datasets. This is due to the effective extraction of both spectral and spatial features and reduced influence of overfitting by stochastic depth. We can also see that although the sample distribution of the Indian pines dataset is extremely uneven, the proposed network still achieves the best average classification accuracy of 75.11%, which indicates its strong robustness because SDRN reduces the redundancy of the model

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TABLE XI
TRAINING AND TESTING TIME ON THREE DATASETS

		INDIAN PINES	PAVIA UNIVERSITY	SALINAS
MLP	Train(s)	160	334	411
	Test(s)	0.09	0.6	0.5
SVM-RBF	Train(s)	2.3	0.7	2.3
	Test(s)	4.3	7.2	22.3
SAE	Train(s)	127	684	903
	Test(s)	2.3	7.2	22.3
3D-CNN	Train(s)	1600	2680	3402
	Test(s)	2.8	11	17
3D-DCNN	Train(s)	161	339	449
	Test(s)	0.38	1.13	2.37
M3D-DCNN	Train(s)	343	668	870
	Test(s)	0.77	1.85	4.83
DR-CNN	Train(s)	6636	7429	17787
	Test(s)	207	105	295
SSRN	Train(s)	2453	2281	7118
	Test(s)	2	5	12
SDRN	Train(s)	955	1949	3122
	Test(s)	3.1	7.6	19.3

through a simple network structure and a training strategy of stochastic depth.

Figs. 6–8 show the false color images of the three HSI datasets, the corresponding ground truth maps, and the classification maps of the best model on three datasets. It can be seen that the classification result diagram is consistent with the classification accuracy shown in Tables VII–IX. Fig. 6 shows that SVM-RBF and MLP classification methods generate relatively more noisy results, which can be attributed to the simplicity of these two models. They cannot fit the data well, resulting in a large number of prediction errors. Of course, the uneven sample distribution of the Indian Pines dataset also makes the classification result of SSRN inferior to that of SDRN, which can mitigate the impact of this problem. From the figures, we can see that although the region-based sampling method is used to avoid the overlap, the proposed SDRN method could obtain classification maps, which are close to the corresponding ground truth. This indicates that SDRN has a better classification capability than other methods.

To verify the robustness of SDRN for different sizes of training data, we selected different proportions of samples for training. For Indian Pines dataset, we selected 2%, 5%, 10%, 15%, and 20% data for training. For the Pavia University and Salinas dataset, the selected proportions for training were 2%, 4%, 6%, 8%, and 10%, respectively. We show the results in Fig. 9. From Fig. 9, we can see that our proposed method gets better performance than other methods. This demonstrates that SDRN can extract more useful features through spectral and spatial learning blocks using the stochastic depth learning strategy. The results also show that our method is robust to different sizes of training data.

To compare with the existing literature, we also conduct the experiments using the popular random sampling strategy on the Indian Pines dataset. Table X shows that although the results of other methods are similar to the existing literature, our proposed method achieves the best performance, which proves the effectiveness of the proposed method.

Finally, we compare the efficiency of all the methods. All methods were implemented using a workstation with an Intel Core i7-8750h CPU, 16 GB memory, and an NVIDIA GeForce GTX 1060 6 GB graphic card. From Table XI, we can see that the time used by the deep learning-based method is longer than the traditional methods. For 3-D-CNN, it consists of different layers of the 3-D CNN, while, for the DR-CNN, it consists of different regions, which need more time to train. Therefore, the time consumption of DR-CNN is longer than 3-D-CNN. For the SSRN, due to the spectral and spatial residual modules, the consumption time is also longer than 3-D-CNN. Compared with SSRN, SDRN adopts the stochastic depth strategy in the training model, which greatly reduces the training time.

VI. CONCLUSION

This article introduces an SDRN for HSI classification. The proposed network consists of spectral and spatial feature extraction modules, which allow the effective extraction of joint spectral and spatial features. A stochastic depth training method is adopted to improve the efficiency of the training. It randomly skips some residual blocks during training but restores the full network during testing. Through such a simple idea, the forward and backpropagation is speeded up, and higher classification accuracy is obtained. On three commonly used HSI datasets, SDRN has achieved higher classification accuracy than the state-of-the-art methods on real data. Moreover, our proposed method greatly shortened the training time and made the training more efficient compared with SSRN with similar structures. In the future, we will explore more deep learning methods to reduce the training time.

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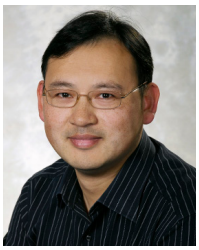
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